Ilya Hsieh-Jarov

UBC Engineering Physics '29

236 334-9425 | ijarov94@gmail.com | LinkedIn | Portfolio | GitHub

Education

The University of British Columbia | Engineering Physics — Vancouver, BC

Cumulative Average: ***85.2%*** | ***Dean's List*** — *Expected 2029*

Relevant Coursework: ENPH 259 (Electronics), ENPH 253 (Robotics), ENPH 353 (Robotics & ML), CPEN 221B

Technical Skills

Software & Control: Python, C, Java, MATLAB, ROS 2, OpenCV, PyTorch, PID Control, Linux, Git

Hardware: Digital Logic, Op-Amp & BJT Circuits, KiCad PCB Design, ESP32, Oscilloscope, Logic Analyzer

CAD & Fabrication: SolidWorks, Onshape, 3D Printing, Machining, Welding

Projects

ROS 2 Robotics & Machine Learning — ENPH 353 — Sep 2026 – Present

University Robotics & Machine Learning Coursework — *Vancouver, BC*

- Competition robot (in progress): developing PyTorch models to locate and read clue boards with a CNN and to drive from camera images using imitation and reinforcement learning.
- Trained a PyTorch policy network with the cross-entropy method to balance a Gazebo cart-pole, and a tabular Q-learning line follower in ROS 2/Gazebo, tracking episode reward in TensorBoard.
- Trained a 36-class Keras CNN on synthetic licence-plate characters (100% validation, 79% test accuracy); a confusion matrix showed that most test errors were misread digits.
- Wrote a ROS 2 Python node for a Gazebo robot: OpenCV threshold and row scan locate the line, a proportional controller publishes Twist steering commands, and it slows to recover from line loss.

Autonomous Robot — ENPH 253 — May 2026 – Aug 2026

University Robotics Design Project — *Vancouver, BC*

- Designed a tank-steer drivetrain and a rack-and-pinion, worm-gear arm in Onshape CAD.
- Sized a 1:25 worm-gear gripper to self-lock; fixed gear jamming with a limit switch.
- Integrated encoders, line sensors, actuators and an SPI ADC (extra input pins): followed the full tape course at 15 cm/s and up a 7° incline, left the track for a task, then relocated the line and resumed.
- Removed bounce from wheel-encoder signals with an RC low-pass filter and Schmitt trigger; designed a coil-based metal detector circuit in KiCad and laid out its PCB.
- Trained a 300+ photo vision model for target detection (80% accuracy) on a Raspberry Pi Zero W.

Closed-Loop Motor Speed Controller — ENPH 259 — Sep 2025 – Dec 2025

University Electronics Lab Coursework — *Vancouver, BC*

- Built a closed-loop motor speed controller (encoder, counter, latch, R-2R DAC, op-amp PI amplifier, BJT driver); speed tracked the setpoint linearly ($R^2 = 0.9996$) up to ∼4200 RPM.
- Isolated faults stage by stage with an oscilloscope and logic analyzer (bent latch pin, miswired counters); in an end-of-course test, traced a swapped R-2R ladder by comparing its input and output signals.

Experience

Teaching Assistant — APSC 160 (Intro to Computation in Engineering) — Sep 2026 – Present

The University of British Columbia — *Vancouver, BC*

- Lead weekly C/Arduino labs for 30 first-year students, covering pointers, memory, and algorithms.
- Debug student code live, tracing logic and memory errors to their root cause.

AI Training & Evaluation — Turing (Contract for Google Gemini) — May 2025 – Jan 2026

Remote — *Palo Alto, CA*

- Reviewed Gemini's STEM responses, identifying and explaining reasoning and calculation errors.